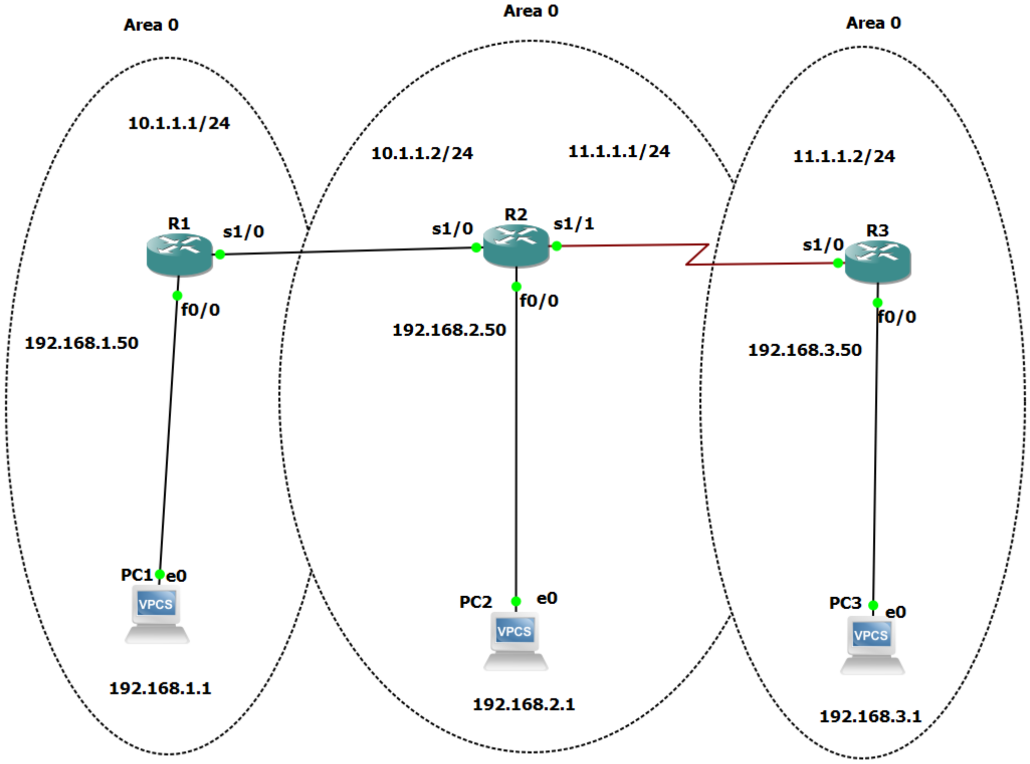
Practical No 6

AIM: OSPF Implementation

1. Implement Single-Area OSPFv2

Step 1: To create a network take 3 routers and 3 PC's



Step 2: Configure PC:

PC1:

PC1> ip 192.168.1.1 255.255.255.0 gateway 192.168.1.50

PC1> sh ip

```
PC1> ip 192.168.1.1 255.255.255.0 gateway 192.168.1.50
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0 gateway 192.168.1.50

PC1> sh ip

NAME         : PC1[1]
IP/MASK      : 192.168.1.1/24
GATEWAY      : 192.168.1.50
DNS          :
MAC          : 00:50:79:66:68:00
LPORT        : 10024
RHOST:PORT   : 127.0.0.1:10025
MTU:         : 1500
```

PC2:

PC2> ip 192.168.2.1 255.255.255.0 gateway 192.168.2.50

PC2> sh ip

```
PC2> ip 192.168.2.1 255.255.255.0 gateway 192.168.2.50
Checking for duplicate address...
PC1 : 192.168.2.1 255.255.255.0 gateway 192.168.2.50

PC2> sh ip

NAME        : PC2[1]
IP/MASK     : 192.168.2.1/24
GATEWAY     : 192.168.2.50
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 10026
RHOST:PORT  : 127.0.0.1:10027
MTU:        : 1500
```

PC3:

PC3> ip 192.168.3.1 255.255.255.0 gateway 192.168.3.50

PC3> sh ip

```
PC3> ip 192.168.3.1 255.255.255.0 gateway 192.168.3.50
Checking for duplicate address...
PC1 : 192.168.3.1 255.255.255.0 gateway 192.168.3.50

PC3> sh ip

NAME        : PC3[1]
IP/MASK     : 192.168.3.1/24
GATEWAY     : 192.168.3.50
DNS         :
MAC         : 00:50:79:66:68:02
LPORT       : 10028
RHOST:PORT  : 127.0.0.1:10029
MTU:        : 1500
```

Step 3: Configure IP Address in Router:

R1:

enable

configure terminal

int f0/0

ip address 192.168.1.50 255.255.255.0

no shutdown

exit

int s1/0

ip address 10.1.1.1 255.255.255.0

no keepalive

no shutdown

exit

end

write memory

no keepalive will make IOS stop waiting for L2 keepalive packets, so Protocol may stay **up**.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int  f0/0
R1(config-if)#ip address 192.168.1.50 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#int  s1/0
R1(config-if)#ip address 10.1.1.1 255.255.255.0
R1(config-if)#no keepalive
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#end
R1#write memory
Building configuration...
[OK]
```

R2

en

conf t

int f0/0

ip add 192.168.2.50 255.255.255.0

no shut

exit

int s1/0

ip add 10.1.1.2 255.255.255.0

no shut

exit

int s1/1

ip add 11.1.1.1 255.255.255.0

no shut

exit

end

wr

```
R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)# ip add 192.168.2.50 255.255.255.0
R2(config-if)# no shut
R2(config-if)# exit
R2(config)#int s1/0
R2(config-if)#ip add 10.1.1.2 255.255.255.0
R2(config-if)# no shut
R2(config-if)# exit
R2(config)#int s1/1
R2(config-if)# ip add 11.1.1.1 255.255.255.0
R2(config-if)# no shut
R2(config-if)# exit
R2(config)#end
R2#wr
```

R3

en

conf t

int f0/0

ip add 192.168.3.50 255.255.255.0

no shut

exit

int s1/0

ip add 11.1.1.2 255.255.255.0

no shut

exit

end

wr

```
R3#en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#int f0/0
R3(config-if)# ip add 192.168.3.50 255.255.255.0
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#int s1/0
R3(config-if)# ip add 11.1.1.2 255.255.255.0
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#end
R3#wr
```

Step 4: Check whether the IP Address assigned is correct or not by using

'do sh ip int br'

R1:

R1(config)#do sh ip int br

```
R1(config)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.1.50    YES manual up                    up
Serial1/0              10.1.1.1        YES manual up                    up
Serial1/1              unassigned      YES unset  administratively down down
Serial1/2              unassigned      YES unset  administratively down down
Serial1/3              unassigned      YES unset  administratively down down
Serial1/4              unassigned      YES unset  administratively down down
Serial1/5              unassigned      YES unset  administratively down down
Serial1/6              unassigned      YES unset  administratively down down
Serial1/7              unassigned      YES unset  administratively down down
```

R2:

R2(config)#do sh ip int br

```
R2(config)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.2.50    YES manual up                    up
Serial1/0              10.1.1.2        YES manual up                    down
Serial1/1              11.1.1.1        YES manual up                    up
Serial1/2              unassigned      YES unset  administratively down down
Serial1/3              unassigned      YES unset  administratively down down
Serial1/4              unassigned      YES unset  administratively down down
Serial1/5              unassigned      YES unset  administratively down down
Serial1/6              unassigned      YES unset  administratively down down
Serial1/7              unassigned      YES unset  administratively down down
```

R3:

R3(config)#do sh ip int br

```
R3(config)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.3.50    YES manual up                    up
Serial1/0              11.1.1.2        YES manual up                    up
Serial1/1              unassigned      YES unset  administratively down down
Serial1/2              unassigned      YES unset  administratively down down
Serial1/3              unassigned      YES unset  administratively down down
Serial1/4              unassigned      YES unset  administratively down down
Serial1/5              unassigned      YES unset  administratively down down
Serial1/6              unassigned      YES unset  administratively down down
Serial1/7              unassigned      YES unset  administratively down down
```

Step 5: Check whether direct connection ping is working in all the routers and PCs:

PC1:

PC1> ping 192.168.1.50

```
PC1> ping 192.168.1.50
84 bytes from 192.168.1.50 icmp_seq=1 ttl=255 time=154.569 ms
84 bytes from 192.168.1.50 icmp_seq=2 ttl=255 time=60.996 ms
84 bytes from 192.168.1.50 icmp_seq=3 ttl=255 time=16.304 ms
84 bytes from 192.168.1.50 icmp_seq=4 ttl=255 time=15.095 ms
84 bytes from 192.168.1.50 icmp_seq=5 ttl=255 time=16.628 ms
```

PC2:

PC2> ping 192.168.2.50

```
PC2> ping 192.168.2.50
84 bytes from 192.168.2.50 icmp_seq=1 ttl=255 time=45.657 ms
84 bytes from 192.168.2.50 icmp_seq=2 ttl=255 time=15.395 ms
84 bytes from 192.168.2.50 icmp_seq=3 ttl=255 time=15.065 ms
84 bytes from 192.168.2.50 icmp_seq=4 ttl=255 time=16.043 ms
84 bytes from 192.168.2.50 icmp_seq=5 ttl=255 time=18.524 ms
```

PC3:

PC3> ping 192.168.3.50

```
PC3> ping 192.168.3.50
84 bytes from 192.168.3.50 icmp_seq=1 ttl=255 time=31.347 ms
84 bytes from 192.168.3.50 icmp_seq=2 ttl=255 time=14.845 ms
84 bytes from 192.168.3.50 icmp_seq=3 ttl=255 time=15.768 ms
84 bytes from 192.168.3.50 icmp_seq=4 ttl=255 time=16.631 ms
84 bytes from 192.168.3.50 icmp_seq=5 ttl=255 time=15.455 ms
```

R1:

R1(config)#do ping 10.1.1.2

```
R1(config)#do ping 10.1.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/28/32 ms
```

R2:

R2(config)#do ping 10.1.1.1

P2(config)#exit

```
R2(config)#do ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/29/44 ms
```

R3:

R3(config)#do ping 11.1.1.2

```
R3(config)#do ping 11.1.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 11.1.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/63/88 ms
```

Direct Connection ping is working successfully. But indirect won't work

because we haven't done any protocol.

So, we will do OSPF in single area.

Step 6: Configure OSPF protocol in all the routers.

R1:

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.1.1.0 0.0.0.255 area 0

```
R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 10.1.1.0 0.0.0.255 area 0
R1(config-router)#ex
```

R2:

R2 OSPF Config

en

conf t

router ospf 1

network 192.168.2.0 0.0.0.255 area 0

network 10.1.1.0 0.0.0.255 area 0

network 11.1.1.0 0.0.0.255 area 0

exit

end

wr

```
R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0
R2(config-router)# network 10.1.1.0 0.0.0.255 area 0
R2(config-router)# network 11.1.1.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#end
R2#wr
Building configuration...
[OK]
```

R3 OSPF Config

en

conf t

router ospf 1

network 192.168.3.0 0.0.0.255 area 0

network 11.1.1.0 0.0.0.255 area 0

exit

end

wr

```
R3#en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)# network 192.168.3.0 0.0.0.255 area 0
R3(config-router)# network 11.1.1.0 0.0.0.255 area 0
R3(config-router)#exit
R3(config)#end
R3#wr
Building configuration...
[OK]
R3#
*Sep 30 11:58:47.179: %SYS-5-CONFIG_I: Configured from console by console
*Sep 30 11:58:48.067: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Sep 30 11:58:49.635: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.50 on Serial1/0 from LOADING to FULL, Loading Don
R3#
```

Step 7: Once OSPF is done enter command 'sh ip route' in all router to check whether OSPF is done properly.

R1:

R1#sh ip route

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.1.0/24 is directly connected, Serial1/0
L        10.1.1.1/32 is directly connected, Serial1/0
      11.0.0.0/24 is subnetted, 1 subnets
O        11.1.1.0 [110/128] via 10.1.1.2, 00:01:53, Serial1/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, FastEthernet0/0
L        192.168.1.50/32 is directly connected, FastEthernet0/0
O     192.168.2.0/24 [110/65] via 10.1.1.2, 00:01:53, Serial1/0
O     192.168.3.0/24 [110/129] via 10.1.1.2, 00:00:32, Serial1/0
R1#
```

R2:

R2#sh ip route

```
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.1.0/24 is directly connected, Serial1/0
L        10.1.1.2/32 is directly connected, Serial1/0
      11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        11.1.1.0/24 is directly connected, Serial1/1
L        11.1.1.1/32 is directly connected, Serial1/1
O     192.168.1.0/24 [110/65] via 10.1.1.1, 00:01:57, Serial1/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/24 is directly connected, FastEthernet0/0
L        192.168.2.50/32 is directly connected, FastEthernet0/0
O     192.168.3.0/24 [110/65] via 11.1.1.2, 00:00:36, Serial1/1
```

R3:

R3#sh ip route

```
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/24 is subnetted, 1 subnets
O        10.1.1.0 [110/128] via 11.1.1.1, 00:00:40, Serial1/0
      11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        11.1.1.0/24 is directly connected, Serial1/0
L        11.1.1.2/32 is directly connected, Serial1/0
O     192.168.1.0/24 [110/129] via 11.1.1.1, 00:00:40, Serial1/0
O     192.168.2.0/24 [110/65] via 11.1.1.1, 00:00:40, Serial1/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected, FastEthernet0/0
L        192.168.3.50/32 is directly connected, FastEthernet0/0
```

Step 8: Enter command 'sh ip protocols' to check which all protocols are applied in our network:

R1:

R1#sh ip protocols

```
R1#sh ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.1.50
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.1.1.0 0.0.0.255 area 0
    192.168.1.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.2.50         110      00:05:35
    192.168.3.50         110      00:04:14
  Distance: (default is 110)
```

R2:

R2#sh ip protocols

```
R2#sh ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.2.50
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.1.1.0 0.0.0.255 area 0
    11.1.1.0 0.0.0.255 area 0
    192.168.2.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.3.50         110      00:04:32
    192.168.1.50         110      00:05:54
  Distance: (default is 110)
```

R3:

R3#sh ip protocols

```
R3#sh ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.3.50
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    11.1.1.0 0.0.0.255 area 0
    192.168.3.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    192.168.2.50         110      00:04:37
    192.168.1.50         110      00:04:37
  Distance: (default is 110)
```

Step 9: Enter command 'sh ip ospf neigbor' to check OSPF Neigbor:

R1:

R1#sh ip ospf neighbor

```
R1#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.2.50      0   FULL/  -        00:00:33    10.1.1.2        Serial1/0
R1#
```

R2:

R2#sh ip ospf neighbor

```
R2#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.3.50      0   FULL/  -        00:00:32    11.1.1.2        Serial1/1
192.168.1.50      0   FULL/  -        00:00:30    10.1.1.1        Serial1/0
```

R3:

R3#sh ip ospf neighbor

```
R3#sh ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.2.50      0   FULL/  -        00:00:38    11.1.1.1        Serial1/0
```

Step 10: Now you can ping any indirect connection because we have doneOSPF on the router

PC1:

PC1> ping 192.168.2.1

```
PC1> ping 192.168.2.1
192.168.2.1 icmp_seq=1 timeout
192.168.2.1 icmp_seq=2 timeout
84 bytes from 192.168.2.1 icmp_seq=3 ttl=62 time=61.552 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=62 time=61.895 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=62 time=47.596 ms

PC1> ping 192.168.3.1
192.168.3.1 icmp_seq=1 timeout
192.168.3.1 icmp_seq=2 timeout
84 bytes from 192.168.3.1 icmp_seq=3 ttl=61 time=90.558 ms
84 bytes from 192.168.3.1 icmp_seq=4 ttl=61 time=92.244 ms
84 bytes from 192.168.3.1 icmp_seq=5 ttl=61 time=95.017 ms
```

PC2> ping 192.168.1.1

```
PC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=62 time=60.719 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=62 time=63.342 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=62 time=62.391 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=62 time=61.825 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=62 time=63.625 ms

PC2> ping 192.168.3.1
84 bytes from 192.168.3.1 icmp_seq=1 ttl=62 time=46.932 ms
84 bytes from 192.168.3.1 icmp_seq=2 ttl=62 time=62.927 ms
84 bytes from 192.168.3.1 icmp_seq=3 ttl=62 time=60.631 ms
84 bytes from 192.168.3.1 icmp_seq=4 ttl=62 time=61.159 ms
84 bytes from 192.168.3.1 icmp_seq=5 ttl=62 time=62.148 ms
```

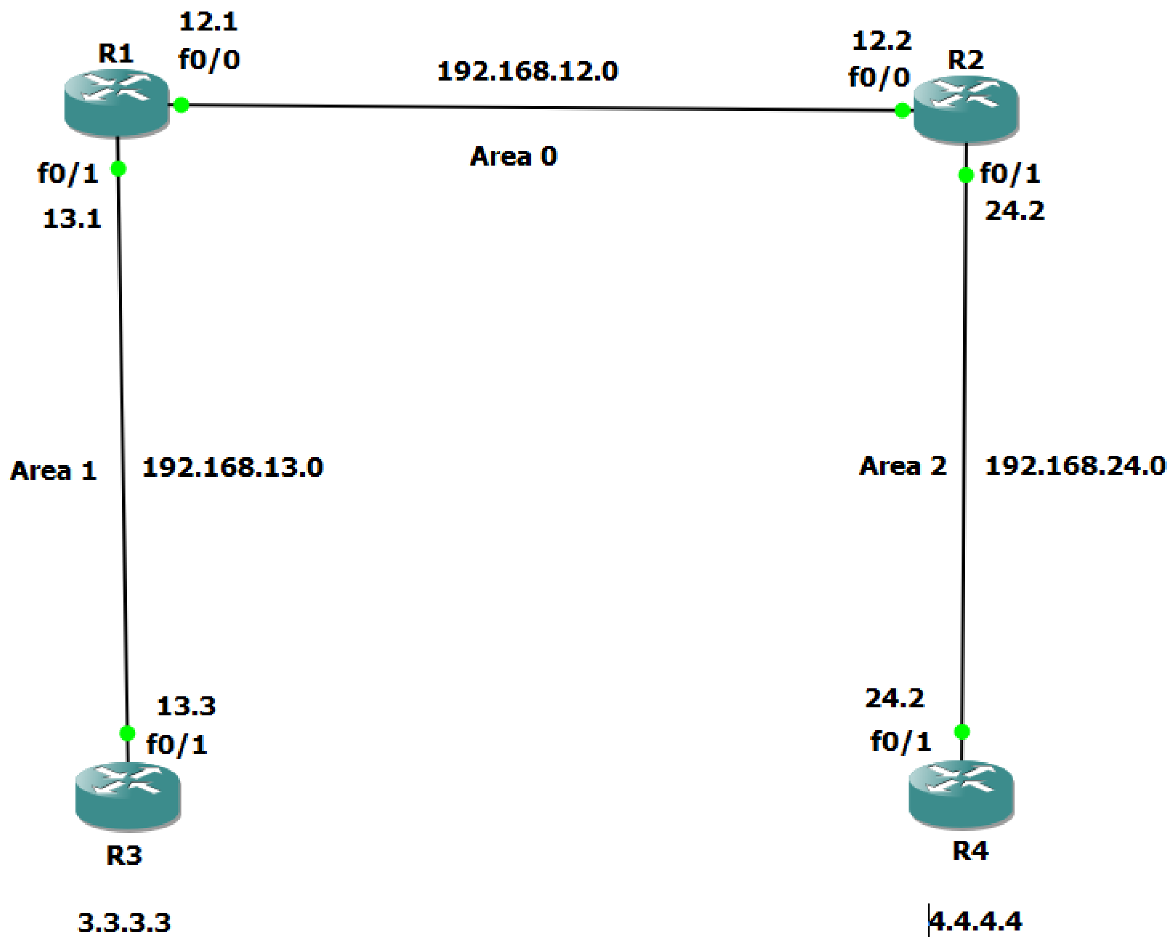
PC3> ping 192.168.1.1

```
PC3> ping 192.168.1.1
192.168.1.1 icmp_seq=1 timeout
192.168.1.1 icmp_seq=2 timeout
84 bytes from 192.168.1.1 icmp_seq=3 ttl=61 time=91.254 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=61 time=93.011 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=61 time=93.907 ms

PC3> ping 192.168.2.1
192.168.2.1 icmp_seq=1 timeout
192.168.2.1 icmp_seq=2 timeout
84 bytes from 192.168.2.1 icmp_seq=3 ttl=62 time=59.712 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=62 time=60.040 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=62 time=62.127 ms
```

2. Implement Multi-Area OSPFv2

Step 1: Take 4 router and make a network as below.



Step 2: Configure all the network as below:

R1:

```
en
conf t
int f0/0
 ip add 192.168.12.1 255.255.255.0
 no shut
 exit
int f0/1
 ip add 192.168.13.1 255.255.255.0
 no shut
 exit
```

end

wr

```
R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)# ip add 192.168.12.1 255.255.255.0
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#int f0/1
R1(config-if)# ip add 192.168.13.1 255.255.255.0
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#end
```

R2

en

conf t

int f0/0

ip add 192.168.12.2 255.255.255.0

no shut

exit

int f0/1

ip add 192.168.24.2 255.255.255.0

no shut

exit

end

wr

```
R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#int f0/0
R2(config-if)# ip add 192.168.12.2 255.255.255.0
R2(config-if)# no shut
R2(config-if)# exit
R2(config)#
R2(config)#int f0/1
R2(config-if)# ip add 192.168.24.2 255.255.255.0
R2(config-if)# no shut
R2(config-if)# exit
R2(config)#
R2(config)#end
R2#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...

*Sep 30 12:54:29.335: %SYS-5-CONFIG_I: Configured from console by console[OK]
R2#
*Sep 30 12:54:31.179: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Sep 30 12:54:31.303: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Sep 30 12:54:32.179: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2#
*Sep 30 12:54:32.303: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
```

R3

```
en
conf t

int f0/1
 ip add 192.168.13.3 255.255.255.0
 no shut
 exit

int loopback0
 ip add 3.3.3.3 255.255.255.255
 no shut
 exit

end
wr
```

```
R3#en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#int f0/1
R3(config-if)# ip add 192.168.13.3 255.255.255.0
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#int loopback0
R3(config-if)# ip add 3.3.3.3 255.255.255.255
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#end
```

R4

en

conf t

int f0/1

 ip add 192.168.24.4 255.255.255.0

 no shut

 exit

int loopback0

 ip add 4.4.4.4 255.255.255.255

 no shut

 exit

end

wr

```
R4#en
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#
R4(config)#int f0/1
R4(config-if)# ip add 192.168.24.4 255.255.255.0
R4(config-if)# no shut
R4(config-if)# exit
R4(config)#
R4(config)#int loopback0
R4(config-if)# ip add 4.4.4.4 255.255.255.255
R4(config-if)# no shut
R4(config-if)# exit
R4(config)#
R4(config)#end
```

Step 3: Now try to ping any router. It won't work because there is no

Protocol applied.

So now we will apply Multi - Area OSPFv2(Area 0, 1, 2).

Configure the system for Multi - Area OSPFv2 as below:

R1 – OSPF Config

en

conf t

router ospf 1

network 192.168.12.0 0.0.0.255 area 0

network 192.168.13.0 0.0.0.255 area 1

exit

end

wr

```
R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network 192.168.12.0 0.0.0.255 area 0
R1(config-router)#network 192.168.13.0 0.0.0.255 area 1
R1(config-router)#exit
R1(config)#end
R1#wr
Building configuration...
[OK]
```

R2 – OSPF Config

en

conf t

router ospf 1

network 192.168.12.0 0.0.0.255 area 0

network 192.168.24.0 0.0.0.255 area 2

exit

end

wr

```
R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)# network 192.168.12.0 0.0.0.255 area 0
R2(config-router)# network 192.168.24.0 0.0.0.255 area 2
R2(config-router)#exit
R2(config)#end
R2#wr
Building configuration...
[OK]
```

R3 – OSPF Config

en

conf t

router ospf 1

network 192.168.13.0 0.0.0.255 area 1

network 3.3.3.3 0.0.0.0 area 1

exit

end

wr

```
R3#en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 192.168.13.0 0.0.0.255 area 1
R3(config-router)#network 3.3.3.3 0.0.0.0 area 1
R3(config-router)#exit
R3(config)#end
R3#wr
Building configuration...
[OK]
```

R4 – OSPF Config

en

conf t

router ospf 1

network 192.168.24.0 0.0.0.255 area 2

network 4.4.4.4 0.0.0.0 area 2

exit

end

wr

```
R4#en
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router ospf 1
R4(config-router)#network 192.168.24.0 0.0.0.255 area 2
R4(config-router)#network 4.4.4.4 0.0.0.0 area 2
R4(config-router)#exit
R4(config)#end
R4#wr
Building configuration...
[OK]
R4#
```

Step 4: Enter the command 'show ip route ospf' to check whether OSPF is successfully configured.

R1#show ip route ospf

```
R1#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      3.0.0.0/32 is subnetted, 1 subnets
O        3.3.3.3 [110/2] via 192.168.13.3, 00:01:22, FastEthernet0/1
      4.0.0.0/32 is subnetted, 1 subnets
O IA     4.4.4.4 [110/3] via 192.168.12.2, 00:00:26, FastEthernet0/0
O IA  192.168.24.0/24 [110/2] via 192.168.12.2, 00:03:32, FastEthernet0/0
```

R2#show ip route ospf

```
R2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      3.0.0.0/32 is subnetted, 1 subnets
O IA     3.3.3.3 [110/3] via 192.168.12.1, 00:01:16, FastEthernet0/0
      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/2] via 192.168.24.4, 00:00:20, FastEthernet0/1
O IA  192.168.13.0/24 [110/2] via 192.168.12.1, 00:03:26, FastEthernet0/0
```

R3#show ip route ospf

```
R3#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
O IA     4.4.4.4 [110/4] via 192.168.13.1, 00:00:17, FastEthernet0/1
O IA  192.168.12.0/24 [110/2] via 192.168.13.1, 00:01:10, FastEthernet0/1
O IA  192.168.24.0/24 [110/3] via 192.168.13.1, 00:01:10, FastEthernet0/1
R3#
```

R4#show ip route ospf

```
R4#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      3.0.0.0/32 is subnetted, 1 subnets
O IA     3.3.3.3 [110/4] via 192.168.24.2, 00:00:24, FastEthernet0/1
O IA  192.168.12.0/24 [110/2] via 192.168.24.2, 00:00:24, FastEthernet0/1
O IA  192.168.13.0/24 [110/3] via 192.168.24.2, 00:00:24, FastEthernet0/1
```

Step 5: To check the neighbor enter 'show ip ospf neighbor' and check the neighbor:

R1#show ip ospf neighbor

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.24.2      1   FULL/BDR        00:00:37    192.168.12.2    FastEthernet0/0
3.3.3.3           1   FULL/BDR        00:00:39    192.168.13.3    FastEthernet0/1
```

R2#show ip ospf neighbor

```
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.13.1      1   FULL/DR         00:00:32    192.168.12.1    FastEthernet0/0
4.4.4.4           1   FULL/BDR        00:00:31    192.168.24.4    FastEthernet0/1
```

R3#show ip ospf neighbor

```
R3#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.13.1      1   FULL/DR         00:00:36    192.168.13.1    FastEthernet0/1
```

R4#show ip ospf neighbor

```
R4#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.24.2      1   FULL/DR         00:00:31    192.168.24.2    FastEthernet0/1
```

As now we have successfully configured and checked that OSPF multi-Area is there in our network. Try pinging any router or loopback from any router.

Step 6:

R1:

tclsh

foreach address {192.168.13.3 192.168.24.4 3.3.3.3 4.4.4.4} {

 ping $address

}

Tclquit

```
R1#tclsh
R1(tcl)#foreach address {192.168.13.3 192.168.24.4 3.3.3.3 4.4.4.4} {
+>(tcl)# ping $address
+>(tcl)#}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.13.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/29/32 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.24.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/61/64 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/31/40 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/61/68 ms
```

R2:

tclsh

foreach address {192.168.13.3 3.3.3.3 4.4.4.4} {

 ping $address

}

tclquit

```
R2#tclsh
R2(tcl)#foreach address {192.168.13.3 3.3.3.3 4.4.4.4} {
+>(tcl)# ping $address
+>(tcl)#}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.13.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/61/64 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/63/68 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/28/32 ms
R2(tcl)#
```

R3:

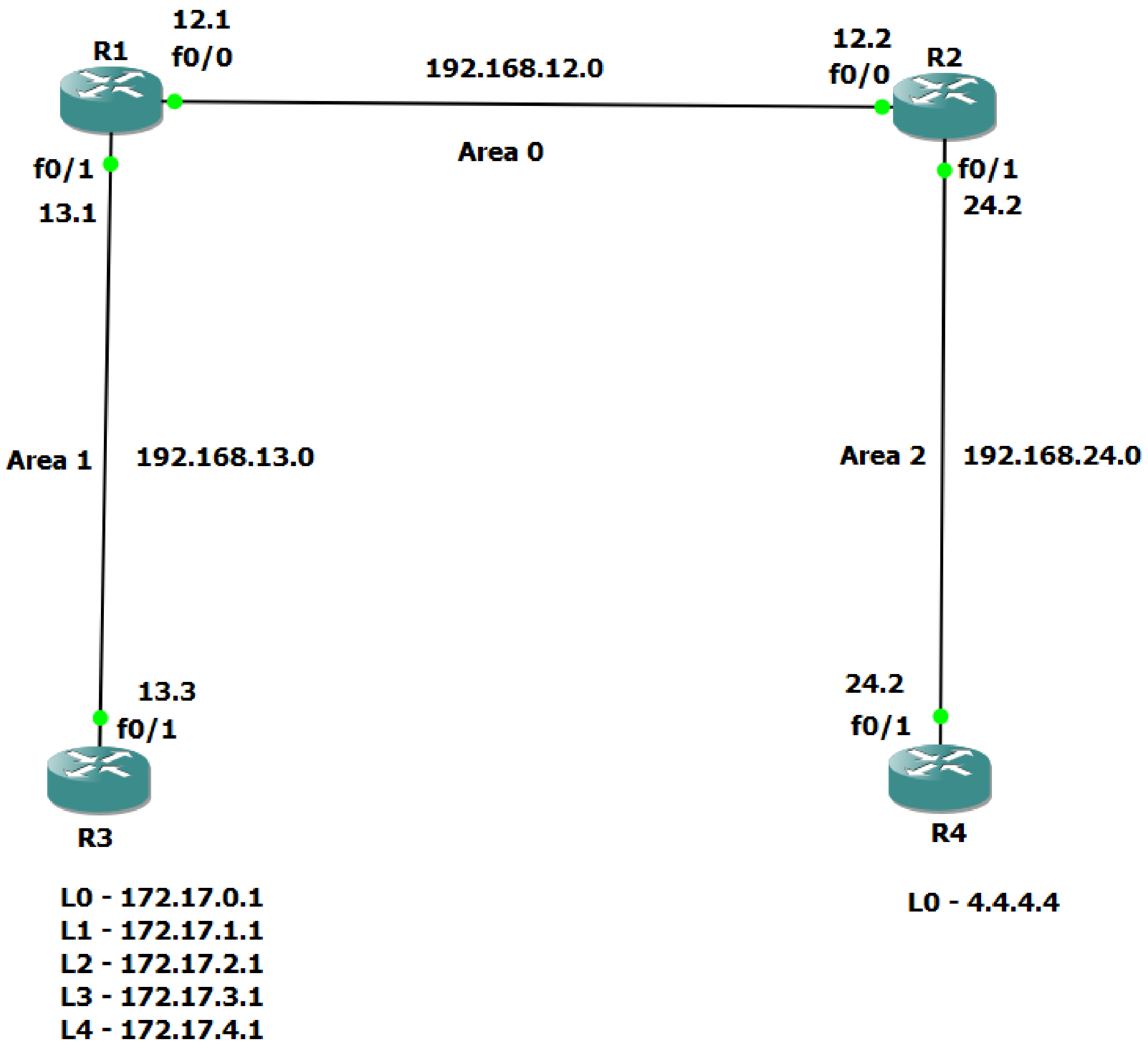
```
R3#tclsh
R3(tcl)#foreach address {192.168.12.2 192.168.24.4 3.3.3.3 4.4.4.4} {
+>(tcl)# ping $address
+>(tcl)#}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/61/72 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.24.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 88/92/96 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 4.4.4.4, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 88/93/96 ms
R3(tcl)#
```

R4:

```
R4#tclsh
R4(tcl)#foreach address {192.168.13.1 192.168.12.1 192.168.13.3 3.3.3.3} {
+>(tcl)# ping $address
+>(tcl)#}
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.13.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/59/64 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/60/68 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.13.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 92/95/100 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 92/95/96 ms
R4(tcl)#
```

3. OSPFv2 Route Summarization and Filtering

Step 1: Follow the same Topology as the Multi - Area OSPFv2.



Step 2: Add more loopbacks to Router 3 and configure the OSPF accordingly.

```
en
conf t

int loopback0
 ip add 172.17.0.1 255.255.255.255
 no shut
 exit

int loopback1
```

```
 ip add 172.17.1.1 255.255.255.255
 no shut
 exit

int loopback2
 ip add 172.17.2.1 255.255.255.255
 no shut
 exit

int loopback3
 ip add 172.17.3.1 255.255.255.255
 no shut
 exit

int loopback4
 ip add 172.17.4.1 255.255.255.255
 no shut
 exit

end
wr
```

```
R3#en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#int loopback0
R3(config-if)# ip add 172.17.0.1 255.255.255.255
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
*Sep 30 13:22:01.351: %SYS-5-CONFIG_I: Configured from console by console
R3(config)#int loopback1
R3(config-if)# ip add 172.17.1.1 255.255.255.255
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#int loopback2
R3(config-if)# ip add 172.17.2.1 255.255.255.255
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#int loopback3
R3(config-if)# ip add 172.17.3.1 255.255.255.255
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#int loopback4
R3(config-if)# ip add 172.17.4.1 255.255.255.255
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#end
R3#wr
Building configuration...
[OK]
R3#
*Sep 30 13:22:02.851: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
*Sep 30 13:22:03.003: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up
*Sep 30 13:22:03.155: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback3, changed state to up
*Sep 30 13:22:03.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback4, changed state to up
```

router ospf 1

network 172.17.0.1 0.0.0.0 area 1

network 172.17.1.1 0.0.0.0 area 1

network 172.17.2.1 0.0.0.0 area 1

network 172.17.3.1 0.0.0.0 area 1

network 172.17.4.1 0.0.0.0 area 1

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)# network 172.17.0.1 0.0.0.0 area 1
R3(config-router)# network 172.17.1.1 0.0.0.0 area 1
R3(config-router)# network 172.17.2.1 0.0.0.0 area 1
R3(config-router)# network 172.17.3.1 0.0.0.0 area 1
R3(config-router)# network 172.17.4.1 0.0.0.0 area 1
R3(config-router)#
```

Step 3: Enter 'show ip route' on R2 and you will see all the loopback of R3.

Because till now we haven't performed any summarization on R1.

R2#show ip route

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/2] via 192.168.24.4, 00:42:30, FastEthernet0/1
      172.17.0.0/22 is subnetted, 1 subnets
O IA     172.17.0.0 [110/3] via 192.168.12.1, 00:00:03, FastEthernet0/0
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.12.0/24 is directly connected, FastEthernet0/0
L        192.168.12.2/32 is directly connected, FastEthernet0/0
O IA  192.168.13.0/24 [110/2] via 192.168.12.1, 00:45:36, FastEthernet0/0
      192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.24.0/24 is directly connected, FastEthernet0/1
L        192.168.24.2/32 is directly connected, FastEthernet0/1
```

Step 4: So now we will perform summarization on R1

router ospf 1

area 1 range 172.17.0.0 255.255.252.0

end

show ip route

```
R1(config)#router ospf 1
R1(config-router)#area 1 range 172.17.0.0 255.255.252.0
R1(config-router)#end
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
O IA     4.4.4.4 [110/3] via 192.168.12.2, 00:11:19, FastEthernet0/0
      172.17.0.0/16 is variably subnetted, 7 subnets, 3 masks
O        172.17.0.0/21 is a summary, 00:01:09, Null0
O        172.17.0.0/22 is a summary, 00:01:09, Null0
O        172.17.0.1/32 [110/2] via 192.168.13.3, 00:01:09, FastEthernet0/1
O        172.17.1.1/32 [110/2] via 192.168.13.3, 00:01:09, FastEthernet0/1
O        172.17.2.1/32 [110/2] via 192.168.13.3, 00:01:09, FastEthernet0/1
O        172.17.3.1/32 [110/2] via 192.168.13.3, 00:01:09, FastEthernet0/1
O        172.17.4.1/32 [110/2] via 192.168.13.3, 00:01:09, FastEthernet0/1
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
 --More--
*Sep 30 13:47:08.847: %SYS-5-CONFIG_I: Configured from console by console
C        192.168.12.0/24 is directly connected, FastEthernet0/0
L        192.168.12.1/32 is directly connected, FastEthernet0/0
      192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.13.0/24 is directly connected, FastEthernet0/1
L        192.168.13.1/32 is directly connected, FastEthernet0/1
O IA  192.168.24.0/24 [110/2] via 192.168.12.2, 00:11:19, FastEthernet0/0
```

Step 5: Once again we will go to R2 and enter the command 'show ip route'. Now we have done summarization on R1 so we will see only 2 loopbacks of R3.

R2#show ip route

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      4.0.0.0/32 is subnetted, 1 subnets
O        4.4.4.4 [110/2] via 192.168.24.4, 00:44:31, FastEthernet0/1
      172.17.0.0/16 is variably subnetted, 2 subnets, 2 masks
O IA     172.17.0.0/21 [110/3] via 192.168.12.1, 00:01:54, FastEthernet0/0
O IA     172.17.0.0/22 [110/3] via 192.168.12.1, 00:02:04, FastEthernet0/0
      192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.12.0/24 is directly connected, FastEthernet0/0
L        192.168.12.2/32 is directly connected, FastEthernet0/0
O IA  192.168.13.0/24 [110/2] via 192.168.12.1, 00:47:37, FastEthernet0/0
      192.168.24.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.24.0/24 is directly connected, FastEthernet0/1
 --More--
```

That's how we do summarization.

Step 6: And now you can ping any loopback of R3 from any router.

Just to confirm I have pinged the loopback of R3 via R4.

R4#ping 172.17.3.1

R4#ping 172.17.4.1

R4#ping 172.17.0.1

```
R4#ping 172.17.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 92/96/104 ms
R4#ping 172.17.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.4.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 92/94/100 ms
R4#ping 172.17.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 84/93/96 ms
```

I have pinged the loopback of R3 via R1.

R1# ping 172.17.0.1

R1# ping 172.17.1.1

R1# ping 172.17.2.1

R1# ping 172.17.3.1

R1# ping 172.17.4.1

```
R1#ping 172.17.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/30/32 ms
R1#ping 172.17.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/25/36 ms
R1#ping 172.17.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/29/32 ms
R1#ping 172.17.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/29/40 ms
R1#ping 172.17.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.4.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/30/32 ms
```

I have pinged the loopback of R3 via R2.

R2# ping 172.17.0.1

R2# ping 172.17.1.1

R2# ping 172.17.2.1

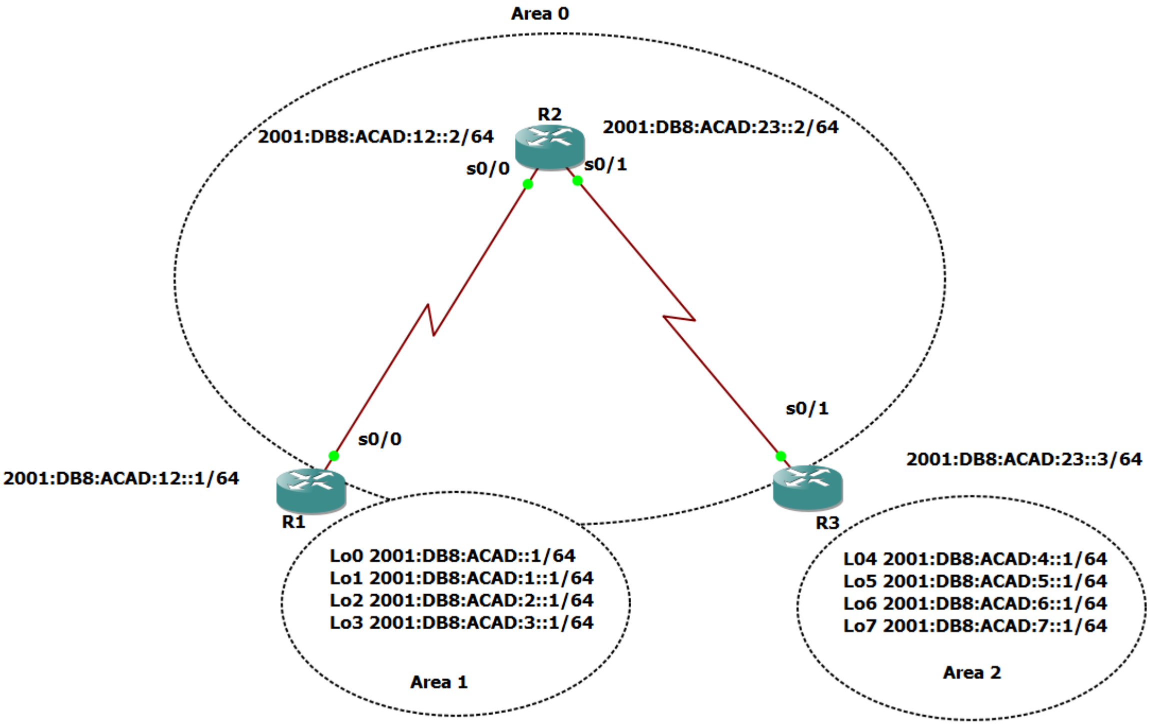
R2# ping 172.17.3.1

R2# ping 172.17.4.1

```
R2#ping 172.17.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/61/68 ms
R2#ping 172.17.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/59/68 ms
R2#ping 172.17.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/61/64 ms
R2#ping 172.17.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/60/64 ms
R2#ping 172.17.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.17.4.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/64/80 ms
```

4. Implement Multiarea OSPFv3

Step 1: Build the topology



Step 2: Configure IP's address and Loopback in all the router according to the topology

We will use IPv6 for OSPF version 3

There's a different command for IPv6 configuration. Follow as below.

R1:

R1 – IPv6 Config

```
en

conf t


int s0/0

 ipv6 address 2001:DB8:ACAD:12::1/64

 no shut

 exit


int lo0

 ipv6 address 2001:DB8:ACAD::1/64

 no shut
```

```
 exit

int lo1
 ipv6 address 2001:DB8:ACAD:1::1/64
 no shut
 exit

int lo2
 ipv6 address 2001:DB8:ACAD:2::1/64
 no shut
 exit

int lo3
 ipv6 address 2001:DB8:ACAD:3::1/64
 no shut
 exit

end
wr
```

```
R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#int s0/0
R1(config-if)# ipv6 address 2001:DB8:ACAD:12::1/64
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#int lo0
R1(config-if)# ipv6 address 2001:DB8:ACAD::1/64
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#int lo1
R1(config-if)# ipv6 address 2001:DB8:ACAD:1::1/64
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#int lo2
R1(config-if)# ipv6 address 2001:DB8:ACAD:2::1/64
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#int lo3
R1(config-if)# ipv6 address 2001:DB8:ACAD:3::1/64
R1(config-if)# no shut
R1(config-if)# exit
R1(config)#
R1(config)#end
R1#
*Mar  1 00:00:15.351: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
*Mar  1 00:00:15.583: %SYS-5-CONFIG_I: Configured from console by console
*Mar  1 00:00:15.967: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback1, changed state to up
R1#
*Mar  1 00:00:16.087: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
R1#
*Mar  1 00:00:16.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback2, changed state to up
*Mar  1 00:00:16.315: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback3, changed state to up
*Mar  1 00:00:17.103: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
```

R2 – IPv6 Config

en

conf t

int s0/0

 ipv6 address 2001:DB8:ACAD:12::2/64

 no shut

 exit

int s0/1

 ipv6 address 2001:DB8:ACAD:23::2/64

 no shut

 exit

int lo8

ipv6 address 2001:DB8:ACAD:8::1/64

no shut

exit

end

wr

```
R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#int s0/0
R2(config-if)# ipv6 address 2001:DB8:ACAD:12::2/64
R2(config-if)# no shut
R2(config-if)# exit
R2(config)#
R2(config)#int s0/1
R2(config-if)# ipv6 address 2001:DB8:ACAD:23::2/64
R2(config-if)# no shut
R2(config-if)# exit
R2(config)#
R2(config)#int lo8
R2(config-if)# ipv6 address 2001:DB8:ACAD:8::1/64
R2(config-if)# no shut
R2(config-if)# exit
R2(config)#
R2(config)#end
R2#wr
Building configuration...

*Mar  1 00:00:56.339: %SYS-5-CONFIG_I: Configured from console by console
*Mar  1 00:00:56.767: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback8, changed state to up
*Mar  1 00:00:57.395: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
*Mar  1 00:00:57.611: %LINK-3-UPDOWN: Interface Serial0/1, changed state to up[OK]
R2#
*Mar  1 00:00:58.483: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
*Mar  1 00:00:58.703: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up
```

R3 – IPv6 Config

en

conf t

int s0/1

ipv6 address 2001:DB8:ACAD:23::3/64

no shut

exit

int lo4

ipv6 address 2001:DB8:ACAD:4::1/64

```
 no shut
 exit

int lo5
 ipv6 address 2001:DB8:ACAD:5::1/64
 no shut
 exit

int lo6
 ipv6 address 2001:DB8:ACAD:6::1/64
 no shut
 exit

int lo7
 ipv6 address 2001:DB8:ACAD:7::1/64
 no shut
 exit

end
wr
```

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#int s0/1
R3(config-if)# ipv6 address 2001:DB8:ACAD:23::3/64
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#int lo4
R3(config-if)# ipv6 address 2001:DB8:ACAD:4::1/64
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#int lo5
R3(config-if)# ipv6 address 2001:DB8:ACAD:5::1/64
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#int lo6
R3(config-if)# ipv6 address 2001:DB8:ACAD:6::1/64
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#int lo7
R3(config-if)# ipv6 address 2001:DB8:ACAD:7::1/64
R3(config-if)# no shut
R3(config-if)# exit
R3(config)#
R3(config)#end
R3#wr
Building configuration...
[OK]
R3#
*Mar  1 00:02:20.879: %SYS-5-CONFIG_I: Configured from console by console
*Mar  1 00:02:21.847: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback4, changed state to up
*Mar  1 00:02:21.863: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback5, changed state to up
*Mar  1 00:02:21.867: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback6, changed state to up
*Mar  1 00:02:21.879: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback7, changed state to up
R3#
*Mar  1 00:02:22.831: %LINK-3-UPDOWN: Interface Serial0/1, changed state to up
R3#
*Mar  1 00:02:23.843: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1, changed state to up
R3#
```

Step 3: Once IP is assigned to all. We have to do IPv6 unicast. And we have to assign router ID to the routers.

R1:

R1

en

conf t

ipv6 unicast-routing

ipv6 router ospf 1

router-id 1.1.1.1

do sh ipv6 ospf

exit

int s0/0

 ipv6 ospf 1 area 0

exit

int lo0

 ipv6 ospf 1 area 0

exit

int lo1

 ipv6 ospf 1 area 0

exit

int lo2

 ipv6 ospf 1 area 0

exit

int lo3

 ipv6 ospf 1 area 0

exit

end

wr

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#
R1(config)#ipv6 router ospf 1
R1(config-rtr)# router-id 1.1.1.1
R1(config-rtr)#do sh ipv6 ospf
 Routing Process "ospfv3 1" with ID 1.1.1.1
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 0. 0 normal 0 stub 0 nssa
 Reference bandwidth unit is 100 mbps

R1(config-rtr)#
*Mar  1 00:07:34.527: %OSPFv3-4-NORTRID: OSPFv3 process 1 could not pick a router-id,
please configure manually
```

R2

```
en
conf t
ipv6 unicast-routing

ipv6 router ospf 1
 router-id 2.2.2.2
do sh ipv6 ospf

exit

int s0/0
 ipv6 ospf 1 area 0
exit

int s0/1
 ipv6 ospf 1 area 0
exit

int lo8
 ipv6 ospf 1 area 0
exit

end
wr
```

```
R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ipv6 unicast-routing
R2(config)#
R2(config)#ipv6 router ospf 1
R2(config-rtr)# router-id 2.2.2.2
R2(config-rtr)#do sh ipv6 ospf
 Routing Process "ospfv3 1" with ID 2.2.2.2
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 0. 0 normal 0 stub 0 nssa
 Reference bandwidth unit is 100 mbps

R2(config-rtr)#
*Mar  1 00:09:54.167: %OSPFv3-4-NORTRID: OSPFv3 process 1 could not pick a router-id,
please configure manually
```

R3

en

conf t

ipv6 unicast-routing

ipv6 router ospf 1

 router-id 3.3.3.3

do sh ipv6 ospf

exit

int s0/1

 ipv6 ospf 1 area 0

exit

int lo4

 ipv6 ospf 1 area 0

exit

int lo5

 ipv6 ospf 1 area 0

exit

int lo6

 ipv6 ospf 1 area 0

exit

int lo7

 ipv6 ospf 1 area 0

exit

end

wr

```
R3#en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ipv6 unicast-routing
R3(config)#
R3(config)#ipv6 router ospf 1
R3(config-rtr)# router-id 3.3.3.3
R3(config-rtr)#do sh ipv6 ospf
 Routing Process "ospfv3 1" with ID 3.3.3.3
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 0. 0 normal 0 stub 0 nssa
 Reference bandwidth unit is 100 mbps

R3(config-rtr)#
*Mar  1 00:09:33.643: %OSPFv3-4-NORTRID: OSPFv3 process 1 could not pick a router-id,
please configure manually
```

Step 4: Now we will configure multi-area OSPFv3 in all the router

R1

en

conf t

ipv6 unicast-routing

ipv6 router ospf 1

 router-id 1.1.1.1

exit

int lo0

 ipv6 ospf 1 area 1

 ipv6 ospf network point-to-point

```
exit

int lo1
 ipv6 ospf 1 area 1
 ipv6 ospf network point-to-point
exit

int lo2
 ipv6 ospf 1 area 1
 ipv6 ospf network point-to-point
exit

int lo3
 ipv6 ospf 1 area 1
 ipv6 ospf network point-to-point
exit

int s0/0
 ipv6 ospf 1 area 0
exit

end
wr
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#ipv6 unicast-routing
R1(config)#ipv6 router ospf 1
R1(config-rtr)# router-id 1.1.1.1
R1(config-rtr)#exit
R1(config)#
R1(config)#int lo0
R1(config-if)# ipv6 ospf 1 area 1
R1(config-if)# ipv6 ospf network point-to-point
R1(config-if)#exit
R1(config)#
R1(config)#int lo1
R1(config-if)# ipv6 ospf 1 area 1
R1(config-if)# ipv6 ospf network point-to-point
R1(config-if)#exit
R1(config)#
R1(config)#int lo2
R1(config-if)# ipv6 ospf 1 area 1
R1(config-if)# ipv6 ospf network point-to-point
R1(config-if)#exit
R1(config)#
R1(config)#int lo3
R1(config-if)# ipv6 ospf 1 area 1
R1(config-if)# ipv6 ospf network point-to-point
R1(config-if)#exit
R1(config)#
R1(config)#int s0/0
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)#exit
R1(config)#
R1(config)#end
```

R2

en

conf t

ipv6 unicast-routing

ipv6 router ospf 1

 router-id 2.2.2.2

exit

int s0/0

ipv6 ospf 1 area 0

exit

int s0/1

ipv6 ospf 1 area 0

exit

int lo8

ipv6 ospf 1 area 0

ipv6 ospf network point-to-point

exit

end

wr

```
R2(config)#int s0/0
R2(config-if)# ipv6 ospf 1 area 0
R2(config-if)#exit
R2(config)#
R2(config)#int s0/1
R2(config-if)# ipv6 ospf 1 area 0
R2(config-if)#exit
R2(config)#
R2(config)#int lo8
R2(config-if)# ipv6 ospf 1 area 0
R2(config-if)# ipv6 ospf network point-to-point
R2(config-if)#exit
R2(config)#
R2(config)#end
R2#wr
Building configuration...
[OK]
R2#
```

R3

en

conf t

ipv6 unicast-routing

```
ipv6 router ospf 1
 router-id 3.3.3.3
exit

int lo4
 ipv6 ospf 1 area 2
 ipv6 ospf network point-to-point
exit

int lo5
 ipv6 ospf 1 area 2
 ipv6 ospf network point-to-point
exit

int lo6
 ipv6 ospf 1 area 2
 ipv6 ospf network point-to-point
exit

int lo7
 ipv6 ospf 1 area 2
 ipv6 ospf network point-to-point
exit

int s0/1
 ipv6 ospf 1 area 0
exit

end
wr
```

```
R3(config)#int lo4
R3(config-if)# ipv6 ospf 1 area 2
R3(config-if)# ipv6 ospf network point-to-point
R3(config-if)#exit
R3(config)#
R3(config)#int lo5
R3(config-if)# ipv6 ospf 1 area 2
R3(config-if)# ipv6 ospf network point-to-point
R3(config-if)#exit
R3(config)#
R3(config)#int lo6
R3(config-if)# ipv6 ospf 1 area 2
R3(config-if)# ipv6 ospf network point-to-point
R3(config-if)#exit
R3(config)#
R3(config)#int lo7
R3(config-if)# ipv6 ospf 1 area 2
R3(config-if)# ipv6 ospf network point-to-point
R3(config-if)#exit
R3(config)#
R3(config)#int s0/1
R3(config-if)# ipv6 ospf 1 area 0
R3(config-if)#exit
R3(config)#
R3(config)#end
R3#wr
```

Step 5: Use the show ipv6 protocols command to verify multi-area OSPFv3 status.

do sh ipv6 protocols

```
R1(config)#do sh ipv6 protocols
IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "static"
IPv6 Routing Protocol is "ospf 1"
  Interfaces (Area 0):
    Serial0/0
  Interfaces (Area 1):
    Loopback3
    Loopback2
    Loopback1
    Loopback0
  Redistribution:
    None
R1(config)#
```

do sh ipv6 protocols

```
R2(config)#do sh ipv6 protocols
IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "static"
IPv6 Routing Protocol is "ospf 1"
  Interfaces (Area 0):
    Loopback8
    Serial0/1
    Serial0/0
  Redistribution:
    None
R2(config)#
```

do sh ipv6 protocols

```
R3(config)#do sh ipv6 protocols
IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "static"
IPv6 Routing Protocol is "ospf 1"
  Interfaces (Area 0):
    Serial0/1
  Interfaces (Area 2):
    Loopback7
    Loopback6
    Loopback5
    Loopback4
  Redistribution:
    None
```

Step 6: Use the 'show ipv6 ospf' command to verify configurations.

R1#show ipv6 ospf

```
R1#show ipv6 ospf
 Routing Process "ospfv3 1" with ID 1.1.1.1
 It is an area border router
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 Reference bandwidth unit is 100 mbps
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        SPF algorithm executed 5 times
        Number of LSA 16. Checksum Sum 0x08E466
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
    Area 1
        Number of interfaces in this area is 4
        SPF algorithm executed 2 times
        Number of LSA 13. Checksum Sum 0x08CFEE
        Number of DCbitless LSA 0
```

R2#show ipv6 ospf

```
R2#show ipv6 ospf
 Routing Process "ospfv3 1" with ID 2.2.2.2
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 Reference bandwidth unit is 100 mbps
    Area BACKBONE(0)
        Number of interfaces in this area is 3
        SPF algorithm executed 3 times
        Number of LSA 19. Checksum Sum 0x0A2D05
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

R3#show ipv6 ospf

```
R3(config)#
R3#show ipv6 ospf
 Routing Process "ospfv3 1" with ID 3.3.3.3
 It is an area border router
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 Reference bandwidth unit is 100 mbps
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        SPF algorithm executed 2 times
        Number of LSA 16. Checksum Sum 0x090943
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
    Area 2
        Number of interfaces in this area is 4
        SPF algorithm executed 2 times
        Number of LSA 13. Checksum Sum 0x04C337
        Number of DCbitless LSA 0
```

Step 7: Verify OSPFv3 neighbors and routing information.

R1:

R1#sh ipv6 ospf neighbor

```
R1#sh ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/  -        00:00:37    6               Serial0/0
```

R2#sh ipv6 ospf neighbor

```
R2#sh ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
3.3.3.3           1   FULL/  -        00:00:39    7               Serial0/1
1.1.1.1           1   FULL/  -        00:00:32    6               Serial0/0
```

R3#sh ipv6 ospf neighbor

```
R3#sh ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/  -        00:00:34    7               Serial0/1
```

Step 8: Check 'show ipv6 route ospf' to see the OSPF configuration

R1#show ipv6 route ospf

```
R1#sh ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/  -        00:00:37    6               Serial0/0
R1#show ipv6 route ospf
IPv6 Routing Table - 17 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
OI  2001:DB8:ACAD:4::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/0
OI  2001:DB8:ACAD:5::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/0
OI  2001:DB8:ACAD:6::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/0
OI  2001:DB8:ACAD:7::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/0
O   2001:DB8:ACAD:8::/64 [110/65]
     via FE80::C002:6CFF:FE94:0, Serial0/0
O   2001:DB8:ACAD:23::/64 [110/128]
     via FE80::C002:6CFF:FE94:0, Serial0/0
```

R2#show ipv6 route ospf

```
R2#sh ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
3.3.3.3           1   FULL/  -        00:00:39    7               Serial0/1
1.1.1.1           1   FULL/  -        00:00:32    6               Serial0/0
R2#show ipv6 route ospf
IPv6 Routing Table - 15 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
OI  2001:DB8:ACAD::/64 [110/65]
     via FE80::C001:68FF:FE20:0, Serial0/0
OI  2001:DB8:ACAD:1::/64 [110/65]
     via FE80::C001:68FF:FE20:0, Serial0/0
OI  2001:DB8:ACAD:2::/64 [110/65]
     via FE80::C001:68FF:FE20:0, Serial0/0
OI  2001:DB8:ACAD:3::/64 [110/65]
     via FE80::C001:68FF:FE20:0, Serial0/0
OI  2001:DB8:ACAD:4::/64 [110/65]
     via FE80::C003:10FF:FE48:0, Serial0/1
OI  2001:DB8:ACAD:5::/64 [110/65]
     via FE80::C003:10FF:FE48:0, Serial0/1
OI  2001:DB8:ACAD:6::/64 [110/65]
     via FE80::C003:10FF:FE48:0, Serial0/1
OI  2001:DB8:ACAD:7::/64 [110/65]
     via FE80::C003:10FF:FE48:0, Serial0/1
```

R3#show ipv6 route ospf

```
R3#show ipv6 route ospf
IPv6 Routing Table - 17 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
OI  2001:DB8:ACAD::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/1
OI  2001:DB8:ACAD:1::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/1
OI  2001:DB8:ACAD:2::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/1
OI  2001:DB8:ACAD:3::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/1
O   2001:DB8:ACAD:8::/64 [110/65]
     via FE80::C002:6CFF:FE94:0, Serial0/1
O   2001:DB8:ACAD:12::/64 [110/128]
     via FE80::C002:6CFF:FE94:0, Serial0/1
```

Step 9: Issue the 'show ipv6 ospf database' command on all routers to check the IPv6 OSPF Database

R1#show ipv6 ospf database

```
R1#sh ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           1   FULL/  -        00:00:37    6               Serial0/0
R1#show ipv6 route ospf
IPv6 Routing Table - 17 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
OI  2001:DB8:ACAD:4::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/0
OI  2001:DB8:ACAD:5::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/0
OI  2001:DB8:ACAD:6::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/0
OI  2001:DB8:ACAD:7::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/0
O   2001:DB8:ACAD:8::/64 [110/65]
     via FE80::C002:6CFF:FE94:0, Serial0/0
O   2001:DB8:ACAD:23::/64 [110/128]
     via FE80::C002:6CFF:FE94:0, Serial0/0
R1#show ipv6 ospf database

            OSPFv3 Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

ADV Router      Age         Seq#        Fragment ID  Link count  Bits
1.1.1.1         503         0x80000002  0            1           B
2.2.2.2         437         0x80000002  0            2           None
3.3.3.3         433         0x80000002  0            1           B

                Inter Area Prefix Link States (Area 0)

ADV Router      Age         Seq#        Prefix
1.1.1.1         574         0x80000001  2001:DB8:ACAD:3::/64
1.1.1.1         574         0x80000001  2001:DB8:ACAD:2::/64
1.1.1.1         574         0x80000001  2001:DB8:ACAD:1::/64
1.1.1.1         574         0x80000001  2001:DB8:ACAD::/64
3.3.3.3         429         0x80000001  2001:DB8:ACAD:7::/64
3.3.3.3         429         0x80000001  2001:DB8:ACAD:6::/64
3.3.3.3         429         0x80000001  2001:DB8:ACAD:5::/64
3.3.3.3         429         0x80000001  2001:DB8:ACAD:4::/64

                Link (Type-8) Link States (Area 0)

ADV Router      Age         Seq#        Link ID    Interface
1.1.1.1         598         0x80000001  6          Se0/0
2.2.2.2         520         0x80000001  6          Se0/0
```

R2#show ipv6 ospf database

```
R2#show ipv6 route ospf
IPv6 Routing Table - 15 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
OI  2001:DB8:ACAD::/64 [110/65]
     via FE80::C001:68FF:FE20:0, Serial0/0
OI  2001:DB8:ACAD:1::/64 [110/65]
     via FE80::C001:68FF:FE20:0, Serial0/0
OI  2001:DB8:ACAD:2::/64 [110/65]
     via FE80::C001:68FF:FE20:0, Serial0/0
OI  2001:DB8:ACAD:3::/64 [110/65]
     via FE80::C001:68FF:FE20:0, Serial0/0
OI  2001:DB8:ACAD:4::/64 [110/65]
     via FE80::C003:10FF:FE48:0, Serial0/1
OI  2001:DB8:ACAD:5::/64 [110/65]
     via FE80::C003:10FF:FE48:0, Serial0/1
OI  2001:DB8:ACAD:6::/64 [110/65]
     via FE80::C003:10FF:FE48:0, Serial0/1
OI  2001:DB8:ACAD:7::/64 [110/65]
     via FE80::C003:10FF:FE48:0, Serial0/1
R2#show ipv6 ospf database

            OSPFv3 Router with ID (2.2.2.2) (Process ID 1)

                Router Link States (Area 0)

ADV Router      Age         Seq#        Fragment ID  Link count  Bits
1.1.1.1         642         0x80000002  0            1           B
2.2.2.2         574         0x80000002  0            2           None
3.3.3.3         570         0x80000002  0            1           B

                Inter Area Prefix Link States (Area 0)

ADV Router      Age         Seq#        Prefix
1.1.1.1         712         0x80000001  2001:DB8:ACAD:3::/64
1.1.1.1         712         0x80000001  2001:DB8:ACAD:2::/64
1.1.1.1         712         0x80000001  2001:DB8:ACAD:1::/64
1.1.1.1         712         0x80000001  2001:DB8:ACAD::/64
3.3.3.3         566         0x80000001  2001:DB8:ACAD:7::/64
3.3.3.3         566         0x80000001  2001:DB8:ACAD:6::/64
3.3.3.3         566         0x80000001  2001:DB8:ACAD:5::/64
3.3.3.3         566         0x80000001  2001:DB8:ACAD:4::/64

                Link (Type-8) Link States (Area 0)
 --More--
```

R3#show ipv6 ospf database

```
R3#show ipv6 route ospf
IPv6 Routing Table - 17 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
OI  2001:DB8:ACAD::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/1
OI  2001:DB8:ACAD:1::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/1
OI  2001:DB8:ACAD:2::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/1
OI  2001:DB8:ACAD:3::/64 [110/129]
     via FE80::C002:6CFF:FE94:0, Serial0/1
O   2001:DB8:ACAD:8::/64 [110/65]
     via FE80::C002:6CFF:FE94:0, Serial0/1
O   2001:DB8:ACAD:12::/64 [110/128]
     via FE80::C002:6CFF:FE94:0, Serial0/1
R3#show ipv6 ospf database

            OSPFv3 Router with ID (3.3.3.3) (Process ID 1)

                Router Link States (Area 0)

ADV Router      Age         Seq#        Fragment ID  Link count  Bits
1.1.1.1         647         0x80000002  0            1           B
2.2.2.2         579         0x80000002  0            2           None
3.3.3.3         573         0x80000002  0            1           B

                Inter Area Prefix Link States (Area 0)

ADV Router      Age         Seq#        Prefix
1.1.1.1         717         0x80000001  2001:DB8:ACAD:3::/64
1.1.1.1         717         0x80000001  2001:DB8:ACAD:2::/64
1.1.1.1         717         0x80000001  2001:DB8:ACAD:1::/64
1.1.1.1         717         0x80000001  2001:DB8:ACAD::/64
3.3.3.3         569         0x80000001  2001:DB8:ACAD:7::/64
3.3.3.3         569         0x80000001  2001:DB8:ACAD:6::/64
3.3.3.3         569         0x80000001  2001:DB8:ACAD:5::/64
3.3.3.3         569         0x80000001  2001:DB8:ACAD:4::/64

                Link (Type-8) Link States (Area 0)
 --More--
```

Reachability Tests

On R1

ping ipv6 2001:DB8:ACAD:12::2   ! R2's Serial0/0

ping ipv6 2001:DB8:ACAD:23::3   ! R3's Serial0/1 via R2

ping ipv6 2001:DB8:ACAD:4::1   ! R3 Loopback4

ping ipv6 2001:DB8:ACAD:5::1   ! R3 Loopback5

ping ipv6 2001:DB8:ACAD:8::1   ! R2 Loopback8

```
R1#ping ipv6 2001:DB8:ACAD:12::2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:12::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/28 ms
R1#ping ipv6 2001:DB8:ACAD:23::3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:23::3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/20 ms
R1#ping ipv6 2001:DB8:ACAD:4::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:4::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/16 ms
R1#ping ipv6 2001:DB8:ACAD:5::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:5::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/16 ms
R1#ping ipv6 2001:DB8:ACAD:8::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:8::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

On R2

Neighbor and remote tests:

ping ipv6 2001:DB8:ACAD:12::1   ! R1's Serial0/0

ping ipv6 2001:DB8:ACAD:23::3   ! R3's Serial0/1

ping ipv6 2001:DB8:ACAD::1   ! R1 Loopback0

ping ipv6 2001:DB8:ACAD:3::1   ! R1 Loopback3

ping ipv6 2001:DB8:ACAD:7::1   ! R3 Loopback7

```
R2#ping ipv6 2001:DB8:ACAD:12::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:12::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/12 ms
R2#ping ipv6 2001:DB8:ACAD:23::3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:23::3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/12 ms
R2#ping ipv6 2001:DB8:ACAD::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
R2#ping ipv6 2001:DB8:ACAD:3::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/16 ms
R2#ping ipv6 2001:DB8:ACAD:7::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:7::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/16 ms
R2#
```

On R3

Neighbor and remote loopback tests:

ping ipv6 2001:DB8:ACAD:23::2 ! R2 Serial0/1

ping ipv6 2001:DB8:ACAD:12::1 ! R1 Serial0/0

ping ipv6 2001:DB8:ACAD::1 ! R1 Loopback0

ping ipv6 2001:DB8:ACAD:1::1 ! R1 Loopback1

ping ipv6 2001:DB8:ACAD:8::1 ! R2 Loopback8

```
R3#ping ipv6 2001:DB8:ACAD:23::2     ! R2 Serial0/1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:23::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/16 ms
R3#ping ipv6 2001:DB8:ACAD:23::2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:23::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/12 ms
R3#ping ipv6 2001:DB8:ACAD:12::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:12::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/20 ms
R3#ping ipv6 2001:DB8:ACAD::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/12 ms
R3#ping ipv6 2001:DB8:ACAD:1::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/16 ms
R3#ping ipv6 2001:DB8:ACAD:8::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:8::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/12 ms
```

Now you have successfully configured multi-area OSPF v3 using IPv6